

EXPERIMENT 23

AIM

Examine process names, paths and parent-child relationships and flag unusual processes using predefined forensic rules.

ALGORITHM

1. Store process PID, name, path and parent process.
2. Inspect each process.
3. Flag executables launched from a temporary directory.
4. Flag Office applications spawning unexpected child processes.
5. Collect triggered rules.
6. Display the process and each triggered rule.
7. Stop.

CODE

```
processes = [
    {
        "pid": 101,
        "name": "chrome.exe",
        "path": r"C:\Program Files\Chrome\chrome.exe",
        "parent": "explorer.exe"
    },
    {
        "pid": 202,
        "name": "update.exe",
        "path": r"C:\Users\alice\AppData\Temp\update.exe",
        "parent": "winword.exe"
    }
]

office_apps = {"winword.exe", "excel.exe"}

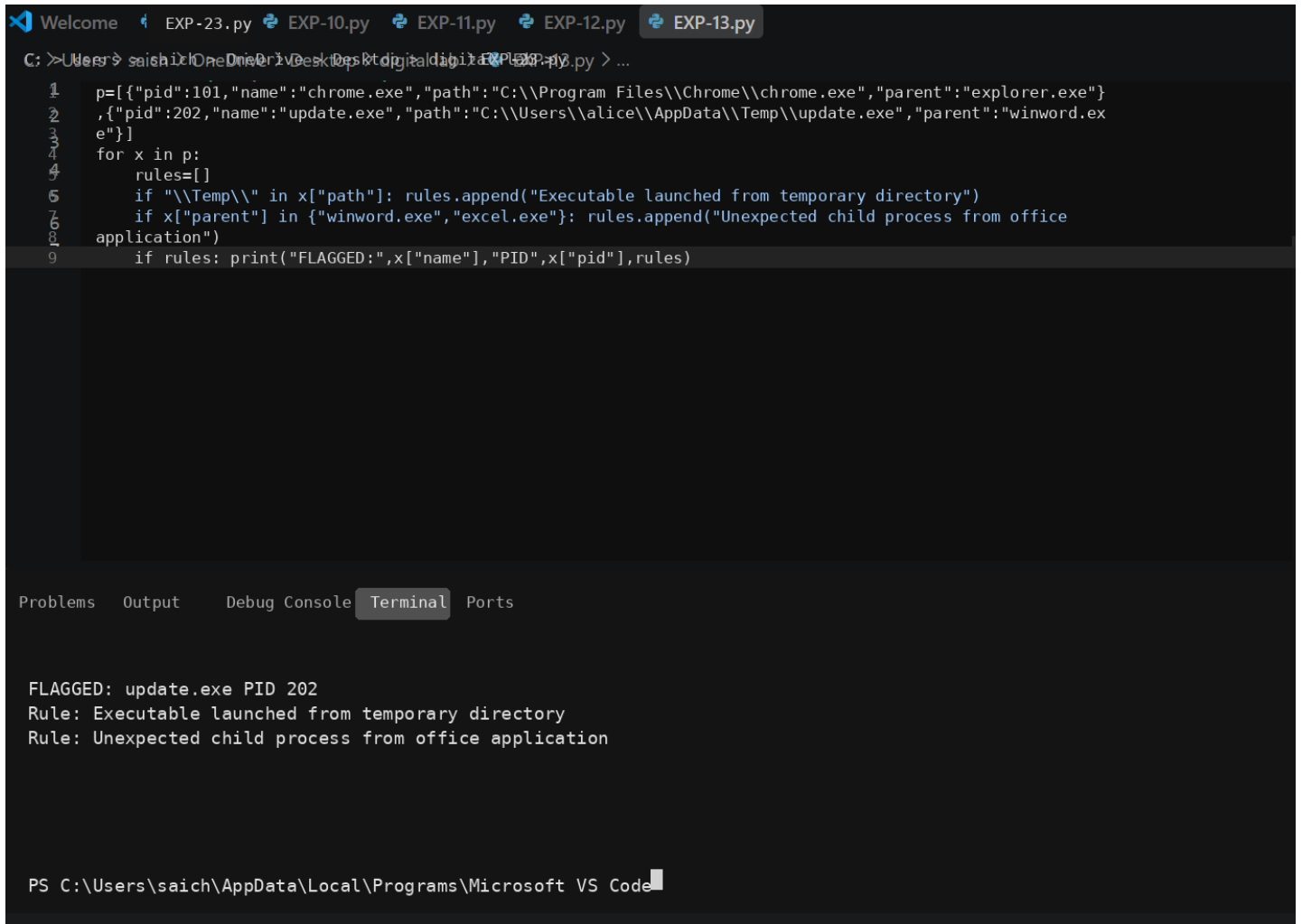
for process in processes:
    rules = []
    path = process["path"].lower()

    if "\\temp\\" in path:
        rules.append("Executable launched from temporary directory")

    if process["parent"].lower() in office_apps:
        rules.append("Unexpected child process from office application")

    if rules:
        print("FLAGGED:", process["name"], "PID", process["pid"])
        for rule in rules:
            print("Rule:", rule)
```

OUTPUT



The screenshot shows a VS Code editor with a file explorer at the top displaying several Python files: EXP-23.py, EXP-10.py, EXP-11.py, EXP-12.py, and EXP-13.py. The main editor window displays the code for EXP-13.py. The code defines a list of processes with their PIDs, names, paths, and parent processes. It then iterates through this list and applies two forensic rules to identify suspicious activity. The terminal output shows the results of these checks, flagging 'update.exe' (PID 202) for two reasons: it was launched from a temporary directory and it is a child process of an office application.

```
1 p=[{"pid":101,"name":"chrome.exe","path":"C:\\Program Files\\Chrome\\chrome.exe","parent":"explorer.exe"}
2 ,{"pid":202,"name":"update.exe","path":"C:\\Users\\alice\\AppData\\Temp\\update.exe","parent":"winword.ex
3 e"}]
4 for x in p:
5     rules=[]
6     if "\\Temp\\" in x["path"]: rules.append("Executable launched from temporary directory")
7     if x["parent"] in {"winword.exe","excel.exe"}: rules.append("Unexpected child process from office
8 application")
9     if rules: print("FLAGGED:",x["name"],"PID",x["pid"],rules)
```

Problems Output Debug Console **Terminal** Ports

```
FLAGGED: update.exe PID 202
Rule: Executable launched from temporary directory
Rule: Unexpected child process from office application
```

PS C:\Users\saich\AppData\Local\Programs\Microsoft VS Code

RESULT

The program flags unusual processes and reports the forensic rule that triggered each alert.